Data-Driven Gradual Preseason Practice Acclimation Ramp Reduces Lower Extremity Strains in the National Football League

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Background: In the National Football League (NFL), approximately 1 in 4 players sustains a lower extremity (LEX) strain each season. Nearly 20% occur in the ~2-week period of training camp at the start of preseason, and LEX strains have a high rate of recurrence.

Purpose: To examine preseason practice schedule strategies during the first week of training camp in the NFL to identify factors associated with lower injury incidence; use findings to implement an actionable, league-wide injury reduction strategy and evaluate effectiveness.

Study Design: Descriptive epidemiology study.

Methods: All players participating during the 2018-2023 seasons were included in this observational study. Injury data from the league-wide electronic medical record were combined with wearable player tracking data and practice schedules. Club practice patterns were analyzed and described and then grouped into strategies. Primary outcomes were hamstring, quadriceps, adductor, and calf LEX strains. Incidence rates per club practice and 95% CIs were calculated.

Results: Between 2018 and 2021, a total of 5 acclimation practice schedule strategies over the first 3 practices were identified from 82 club-seasons: gradual 15-minute ramp (increasing duration 15 minutes each day), <10-minute ramp (increasing duration <10 minutes each day), high-low duration variation (varying duration), stable duration, and reverse ramp (decreasing duration each day). Clubs using the gradual 15-minute ramp, <10-minute ramp, or high-low duration variation had lower training camp LEX strain rates per club-practice compared with the other strategies (Incidence rate ratio = 0.81; 95% CI, 0.68-0.95; *P* = .01). Results were communicated to stakeholders before the 2022 preseason, and a mandatory league-wide gradual practice acclimation ramp was implemented by the NFL and NFL Players Association as part of the Collective Bargaining Agreement, coupled with mandatory club one-on-one education sessions to explain the requirements. League-wide LEX strain incidence during training camp decreased from 288 in 2021 to 215 in 2022 and 186 in 2023. Additionally, recurrent LEX strains in the regular season decreased from 22 per year in 2021 to 11 per year in 2022-2023.

Conclusion: A mandated gradual practice acclimation ramp in training camp coupled with education around strategic load management was associated with reduced LEX strains in the NFL. After the intervention, league-wide LEX strain incidence during training camp decreased by 35%.

Keywords: lower extremity strain; hamstring strain; adductor strain; NFL; football; epidemiology

Lower extremity (LEX) strains, specifically to the hamstring, quadriceps, calf, and adductor, are common injuries across all levels of sports.[2,7,8,15,23,26] LEX strains lead to a wide range of time-loss from participation depending on the location and severity of injury[21] and can contribute a high burden to players and teams.[7,20,21] Furthermore, LEX strains, especially hamstring strains, have been reported to recur frequently.[23,24]

In the US National Football League (NFL), approximately 1 in 4 players sustained a LEX strain each year between 2015 and 2019.[15] The majority of these injuries (69%) led to time-loss from football activities, among which the median number of days missed was 12 days.[15] In terms of cumulative missed participation from football activity, as well as games missed across the NFL, the burden of

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LEX strains is greater than that from anterior cruciate ligament injuries.[20] In the preseason, muscle strains, especially hamstring strains, were previously reported to be the most frequent injury in practices across 10 years among one NFL club.[9] League-wide, nearly 20% of all LEX strains across the NFL occurred during training camp, the 2- to 3-week period at the start of preseason before the first preseason game.[15]

The concept of training load management has been an increasing point of focus in athletic settings over the past decade and is considered a lever toward injury reduction, but the literature is generally inconclusive about whether and how training load influences injury.[3,6,18] Previous studies have identified that "periods of training load intensification," such as during the preseason, may be associated with increased injury risk.[18] Specific training load interventions that may reduce the risk of injury have not been clearly identified,[18] although practice schedule management is hypothesized to be one strategy to manage training load to reduce injury.

The high incidence of LEX strains reported during training camp, which represents <10% of the active NFL season, was identified as an area of opportunity for injury reduction in the NFL.[15] The purpose of this study was to examine training camp practice schedule strategies across NFL clubs to identify practice schedule factors associated with lower LEX strain incidence with the goal of identifying, implementing, and evaluating an actionable, evidence-based, league-wide injury reduction strategy. The hypothesis was that a load management strategy that reduces injuries may be able to be identified and recommended after evaluation of league-wide player tracking data from training camp practices.

METHODS

Study Population and Data Sources

Study approvals were obtained through the NFL Player Scientific and Medical Research Protocol[20] and by the NFL Player Scientific and Medical Research Protocol and the Mount Sinai Institutional Review Board (Pro00060188). All NFL players participating during the 2018 through 2023 seasons were included in this observational study. Injury data were obtained from the league-wide electronic medical record (EMR). Standardized data collection, reporting requirements, and data quality practices across all 32 NFL clubs are described in detail in a previous publication.[4] Club medical staff report details of injury occurrence, treatment, and return to participation and log each team activity in an "activity calendar" within the EMR.

Injury data from the EMR were linked to wearable player tracking data from preseason practices. Tracking data were provided voluntarily by clubs for the 2018-2019 seasons. In the 2020-2021 seasons, clubs were required to submit for analysis any data collected by these devices, and for the 2022-2023 seasons clubs were required to both collect and share data. Clubs collected tracking data from 5 vendors: Catapult Group International, Zebra Technologies, Polar Electro Oy, Kinexon, and STATSports. Player tracking data submitted (ie, duration of participation) were comparable across vendors and included time-stamped sensor data for individual players and the "periods" or drills that occurred during practice. These devices have been validated for metrics such as distance and speed.[1,10,11,13,16,17,27] Duration was defined as the beginning of the first period in which the athlete participated through the end of the last period in which the player appeared and was calculated for each athlete from the raw sensor data for the entire practice.

Study Period

During the 2018-2023 study period, most players reported to preseason training camp on the veteran report date (day 1 of training camp) as required by the NFL–NFL Players Association (NFLPA) Collective Bargaining Agreement (CBA),[22] and the first practice occurred on the day after the veteran report date. Training camp consisted of a 3- to 5-day noncontact acclimation period followed by padded practices, with structure changes during the study period including (1) a cancelled offseason program and modified training camp in 2020 due to the COVID-19 pandemic and (2) a reduction in preseason games (from 4 to 3) and 2 additional days of training camp in 2021 (Figure 1).

For this study, the club practice schedule strategy analysis included the 2018-2021 seasons (preintervention period) and evaluation of the effectiveness of the intervention included the 2022-2023 seasons (postintervention period).

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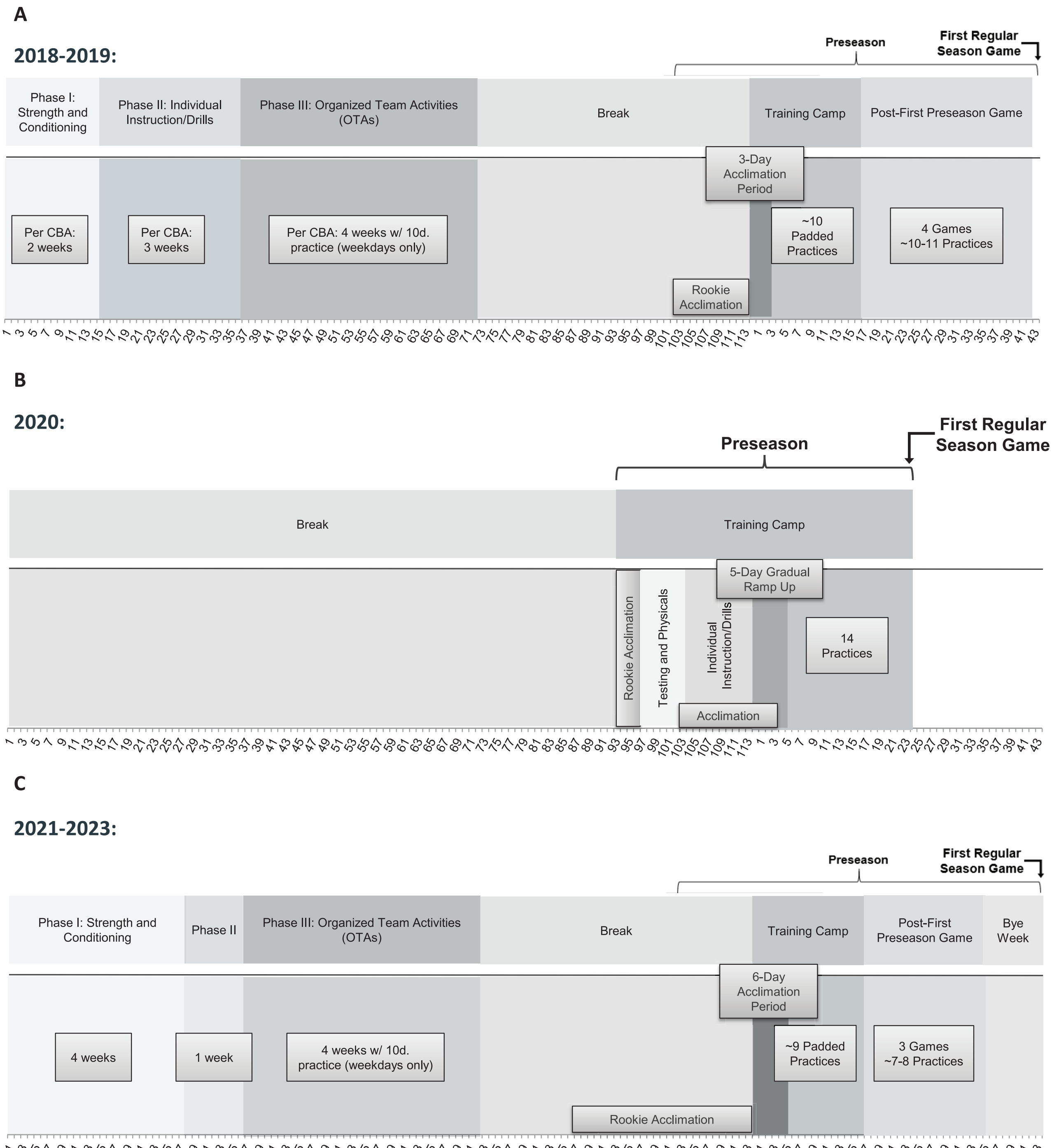


Figure 1. National Football League (NFL) offseason and preseason structure changes, 2018-2023. (A) 2018-2019. (B) 2020. (C) 2021-2023. CBA, Collective Bargaining Agreement.

Exposure: Club Practice Schedule Strategies

Data from wearable player tracking devices were combined with EMR activity calendar data to identify club practices and practice duration (including sessions tagged as “practice,” “walkthrough,” and “conditioning”); sessions tagged as “rehab” were excluded.

Mean daily practice duration for each club was calculated from tracking data for each player who participated in that session. If tracking data were unavailable or <30 players were tracked for a given session, the mean duration metrics were set to “missing” to avoid misrepresenting the full team’s practice experience. To increase accuracy of estimated durations, 3 methods were compared for mean

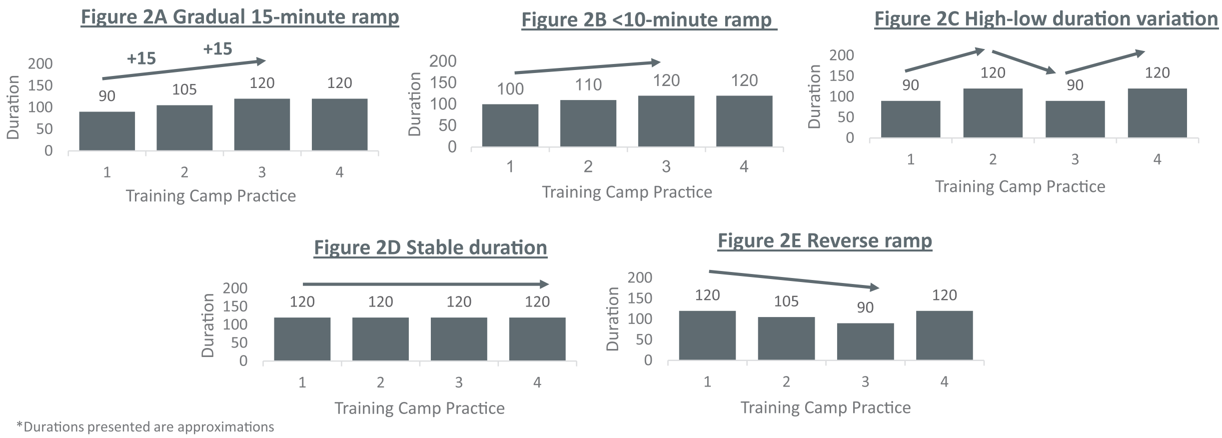


Figure 2. Preseason practice schedule strategies, 2018-2021. (A) Gradual 15-minute ramp: increasing duration with each of the first 3 practices by ~15 minutes; (B) <10-minute ramp: increasing duration with each of the first 3 practices, by <10 minutes; (C) High-low duration variation: alternating pattern of duration, where practice 2 is longer than practices 1 and 3; (D) Stable duration: minimal variation in practice duration; (E) Reverse ramp: decreasing duration with each practice, where practice 1 duration is higher than practices 2 and 3.

duration of practice: mean duration (start to end of practice) and mean active duration (defined by when the sensor is active for the practice session) from player tracking devices, and mean time calculated from start to end time recorded in the EMR activity calendar session.

Club practice schedules for training camp week 1 were individually analyzed by plotting practice duration by day during training camp to describe club frequency and duration of practices. Through the descriptive review, 3 different general strategies were observed across the first week of training camp practices: gradual increase in practice duration, varied practice duration or "high-low" each practice, and gradual decrease in practice duration. The investigators subsequently created 5 categories by assessing practice duration in the first 3 or 4 training camp practices and grouped each club-season into the following strategies (Figure 2) (durations are approximations):

- Gradual 15-minute ramp: increasing duration with each practice by ≥15 minutes
- <10-minute ramp: increasing duration with each practice by <10 minutes
- High-low duration variation: alternating pattern of duration, where practice 2 is longer than practices 1 and 3
- Stable duration: minimal variation in practice duration
- Reverse ramp: decreasing duration with each practice, where practice 1 duration is higher than practices 2 and 3

Preseason practice schedule strategies were independently assigned for each club-season by 2 investigators based on all available data. A third investigator adjudicated any disagreements in strategy assigned. If the calculated mean duration metrics were missing for any practice during training camp week 1, the club-season was excluded from the analysis of preseason practice schedule strategies since the strategy for that club-season could not be definitively determined. Secondary analyses to test the robustness of findings excluded sessions tagged as "walkthrough" or "conditioning" to narrow the exposure definition to full team practices.

Outcomes: LEX Strains and Noncontact Injuries

The primary outcomes were LEX strains to the hamstring, quadriceps, adductor, and calf diagnosed and reported by club medical staffs using NFL-specific "clinical impression codes." All LEX strains were included, regardless of whether they resulted in time-loss from participation. Sensitivity analyses were performed restricting to only LEX strains that resulted in time-loss from participation. Time-loss injuries were defined as an injury that resulted in a player being unable to participate in 100% of their normal repetitions in subsequent activities.

Time-loss noncontact injuries were analyzed as secondary outcomes under the hypothesis that load management strategies such as gradual acclimation may mitigate other soft tissue injuries. These injuries were defined as injuries reported by medical staffs to have occurred by a noncontact or insidious/overuse mechanism.

Injuries were analyzed in 2 time periods: (1) training camp (from veteran report date through the day before first preseason game for all years except 2020, which used days 1-17) and (2) the full preseason period from veteran report date through the day before first regular season practice.

The incidence of recurrent LEX strains in regular season was calculated, defined as an ipsilateral injury to the same muscle after return to full participation from the index LEX strain.

Statistical Analysis

Descriptive statistics, including counts, percentages, medians, and interquartile ranges (IQRs), were calculated for descriptive metrics such as injury incidence and use of different preseason practice strategies. Injury incidence rates, incidence rate ratios (IRR), and 95% CIs were calculated as the number of injuries per club-practice or per 100 practice-minutes using log-Poisson regression models, and corresponding P values were generated.

TABLE 1
Training Camp Lower Extremity Strain and Noncontact Injury Rates by Preseason Practice Strategy, 2018-2021[a]

	Gradual 15-Minute Ramp	<10-Minute Ramp	High-Low Duration Variation	Stable Duration	Reverse Ramp
Unique clubs	14	17	9	13	6
Club seasons	15	27	13	18	9
Training camp practice, min, median (IQR)	1396 (1212-1585)	1396 (1202-1516)	1370 (1300-1536)	1400 (1161-1636)	1407 (1150-1445)
Lower extremity strains					
Rate per club-practice (95% CI)	0.48 (0.39-0.59)	0.55 (0.47-0.63)	0.54 (0.44-0.67)	0.65 (0.56-0.76)	0.66 (0.53-0.83)
Rate per 100 practice-minutes (95% CI)	0.42 (0.34-0.52)	0.44 (0.38-0.51)	0.46 (0.37-0.56)	0.56 (0.48-0.66)	0.57 (0.45-0.72)
Noncontact injuries					
Rate per club-practice (95% CI)	0.49 (0.40-0.61)	0.53 (0.46-0.62)	0.54 (0.44-0.66)	0.68 (0.58-0.79)	0.65 (0.52-0.82)
Rate per 100 practice-minutes (95% CI)	0.44 (0.36-0.54)	0.43 (0.37-0.50)	0.45 (0.37-0.55)	0.59 (0.50-0.69)	0.56 (0.45-0.71)

[a]IQR, interquartile range.

RESULTS

Between 2018 and 2023, a total of 6720 unique players from all 32 NFL Clubs were included in the study. For the preintervention period, 82 club-seasons of player tracking data met inclusion criteria and were included in the practice schedule strategy analysis (2018, n = 12; 2019, n = 20; 2020, n = 24; 2021, n = 26). Twenty-six (24%) club-seasons were excluded from the analysis of practice schedule strategies due to missing practice data or <30 players tracked during 1 or more practices in training camp week 1 (2018, n = 8; 2019, n = 7; 2020, n = 5; 2021, n = 6) (Figure 3). For the analysis of injury trends, complete data were captured for injury counts and rates per practice from all 32 clubs each season and were included.

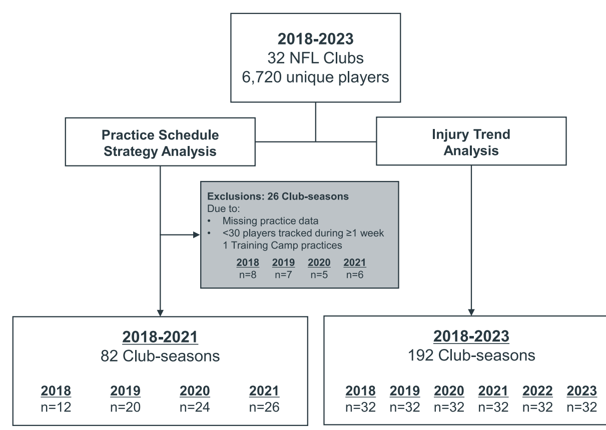


Figure 3. Data inclusion. NFL, National Football League.

Club Practice Schedule Strategies

In 2018-2021, the median training camp week 1 practice duration was 118 minutes (IQR = 103-143). In 2018-2019, most clubs practiced for 4 or 5 consecutive days before taking their first day off (4 days, n = 15 club-seasons, 47%; 5 days, n = 11 club-seasons, 34%), and there was no difference in training camp LEX strain rate between the 4- vs 5-day length before the first day off (0.61 vs 0.65 LEX strains per club-practice; IRR = 0.93; 95% CI, 0.70-1.24). In 2020, due to preseason structural changes, most clubs (n = 19; 79%) practiced for 3 consecutive days before taking their first day off, and in 2021 most clubs (n = 24; 92%) practiced for 4 consecutive days before taking their first day off due to a mandated day off on training camp day 6.

Clubs using a gradual acclimation strategy (gradual 15-minute ramp, <10-minute ramp, or high-low duration variation) had lower LEX strain rates per club-practice during training camp compared with other strategies (IRR = 0.81; 95% CI, 0.68-0.95; P = .01) (Table 1). These findings held after accounting for the number of practice minutes (IRR = 0.78; 95% CI, 0.66-0.92; $P < .01$), and findings were similar when only noncontact time-loss injuries were examined as the outcome (Table 1). LEX strain rates per club-practice for clubs using a gradual acclimation strategy were lower for practices throughout preseason (gradual 15-minute ramp, 0.33; <10-minute ramp, 0.40; high-low duration variation, 0.36; stable duration, 0.46; reverse ramp, 0.47; IRR = 0.81; 95% CI, 0.70-0.93; $P < .01$). Regardless of the practice schedule strategy, clubs had similar total practice minutes across training camp (Table 1), illustrating that with any of these strategies, the same amount of total practice time was attainable.

Intervention: Mandated, League-Wide Gradual Practice Acclimation Ramp and Education

Preliminary results were communicated to club stakeholders, including coaches, front office staff, and medical and performance staff, in mandatory one-on-one club meetings

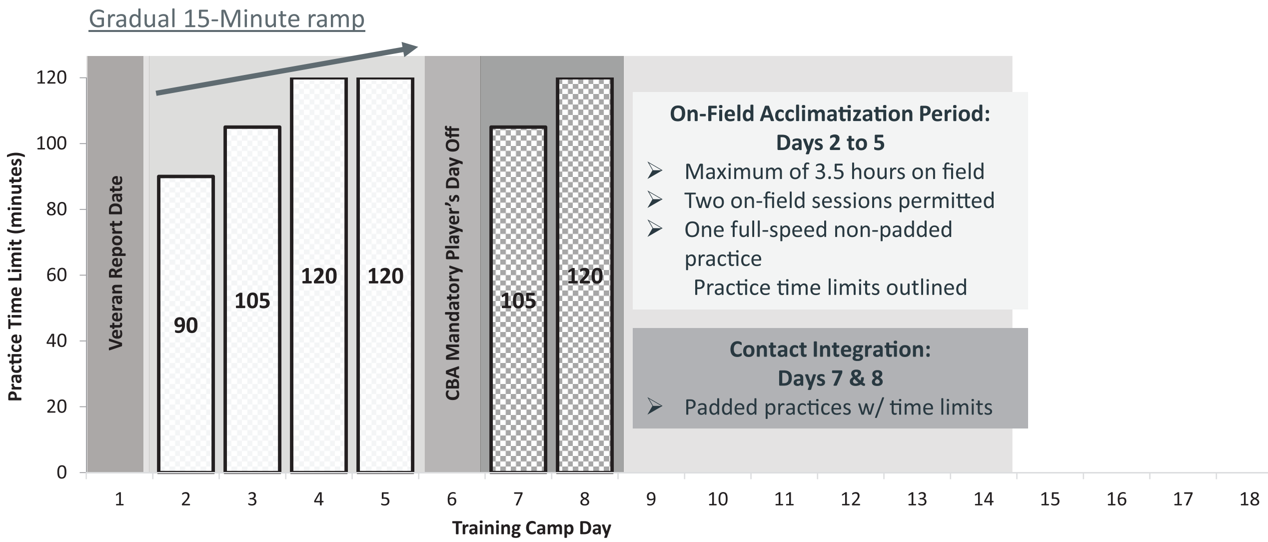


Figure 4. National Football League (NFL) and NFL Players Association (NFLPA) expert recommendations implemented in 2022.

during the 2021 and 2022 offseasons. In 2022, the NFL and NFLPA convened a group of experts to discuss the data. The experts proposed a league-wide gradual practice acclimation ramp limiting practice duration in the first 6 practices to provide guardrails for implementing a load management strategy (Figure 4). These recommendations were implemented for the 2022 training camp through modifications to the NFL-NFLPA CBA. Before the 2022 training camp, the NFL Health and Safety Department and relevant outside expert consultants met with stakeholders from each of the 32 clubs to discuss the data-driven, time-based mandate. These education sessions focused on the importance of load management more broadly, with practice duration being a proxy for strategic management of intensity, distance, and position-specific movements (eg, accelerations, cutting, decelerations).

League-Wide Injury Incidence Before and After the Intervention

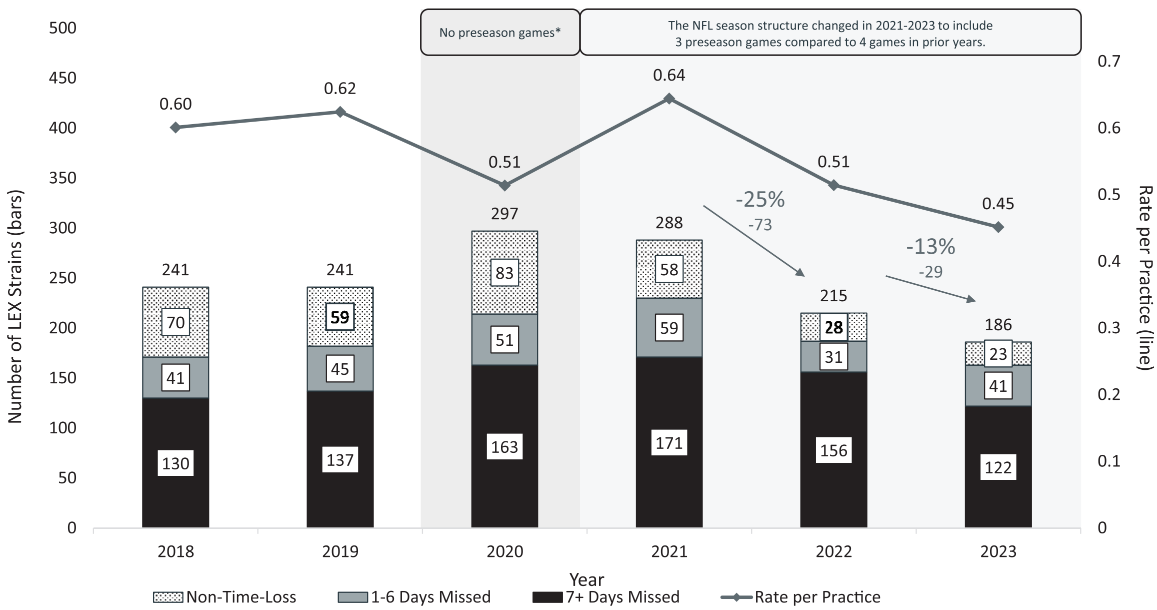
After the first year of the intervention, league-wide LEX strain incidence during training camp decreased by 25%, from 288 in 2021 to 215 in 2022. The mandate was kept in place for a second year, accompanied by additional education, and LEX strain incidence decreased further by an additional 13% to 186 in 2023 (Figure 5), bringing the 2-year decrease postintervention to 35% (102 fewer LEX strains during 2023 training camp vs 2021). When restricted to time-loss LEX strains, incidence decreased by 19% from 2021 (n = 230) to 2022 (n = 187) and by an additional 13% in 2023 (n = 163) (Figure 5). Decreases were largest during the first and second weeks of training camp (2-year decreases of 38% and 33%, respectively) (Figure 6). Compared with 2018-2021 (excluding the year 2020), 21 clubs (66%) had a lower median number of LEX strains in training camp in 2022-2023. These decreases extended beyond LEX strains; 18 clubs (56%) had a lower median number of noncontact time-loss injuries in 2022-2023 vs 2018-2021 (excluding the year 2020).

The benefits of the training camp acclimation ramp requirements extended through the NFL regular season, with league-wide LEX strain incidence from preseason through the last game of the NFL regular season decreasing by 19% from 2021 to 2023 (Figure 7). Although the decreases observed during training camp (the active intervention period) were the largest, decreases were also observed during the post–training camp preseason game period (2021, n = 158; 2022, n = 159; 2023, n = 136) and in the regular season (2021, n = 493; 2022, n = 485; 2023, n = 441). Notably, the median number of recurrent LEX strains in the 2022-2023 regular seasons decreased by 50% compared with 2021 (n = 11 per year vs n = 22, respectively).

DISCUSSION

This study used wearable player tracking data combined with EMR injury data to identify preseason practice schedule factors that were associated with lower injury incidence in the NFL. The results suggest that using a gradual acclimation strategy, such as a gradual increase or high-low variation in practice duration, across the first week of training camp was associated with lower rates of LEX strain and all noncontact injury rates. When these findings were implemented as an intervention before the 2022 season coupled with individual club education sessions, there was a meaningful reduction in league-wide LEX strain and noncontact LEX injury incidence.

The mandated gradual practice acclimation ramp not only was supported by evidence from wearable data from club practice schedules but also was aligned with expert opinion, which guided both the mandate and ongoing education. Given the strength of the data, expert recommendation was to implement this in training camp week 1 and



*No preseason games were played in 2020 and all offseason and preseason activities occurred during a period of ~6 weeks prior to regular season games beginning (See Figure 1). For 2020, only LEX strains from the gradual ramp and contact integration periods are included.

Figure 5. Training camp lower extremity (LEX) strains in practices and other activities, 2018-2023.

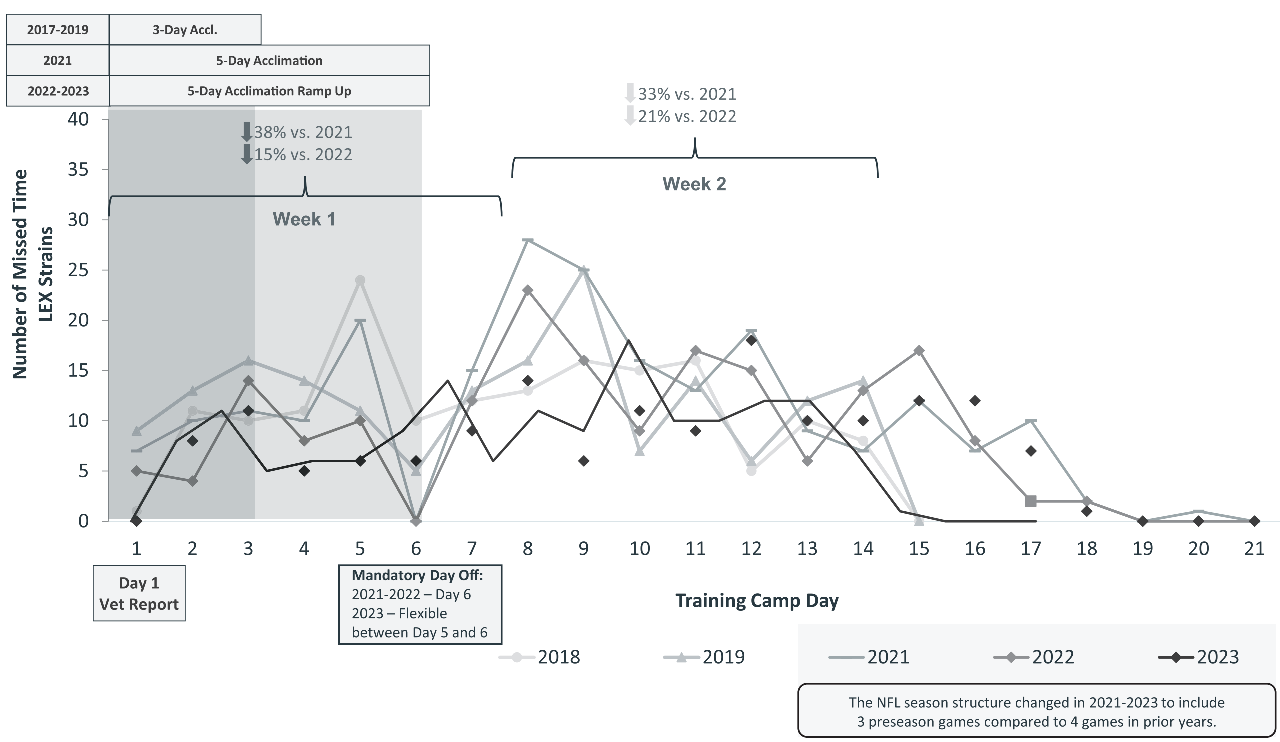


Figure 6. Missed time lower extremity (LEX) strains by training camp day in practices and other activities, 2018-2023.

again in week 2 as a "contact integration re-ramp" once contact practices started, reducing the maximum practice duration by 15 minutes for the fifth practice and then increasing duration by 15 minutes for the sixth practice of the contact integration portion of preseason as well, before resuming CBA maximum practice durations. This extrapolation of findings from the first 4 practices of training camp was supported by expert recommendation that

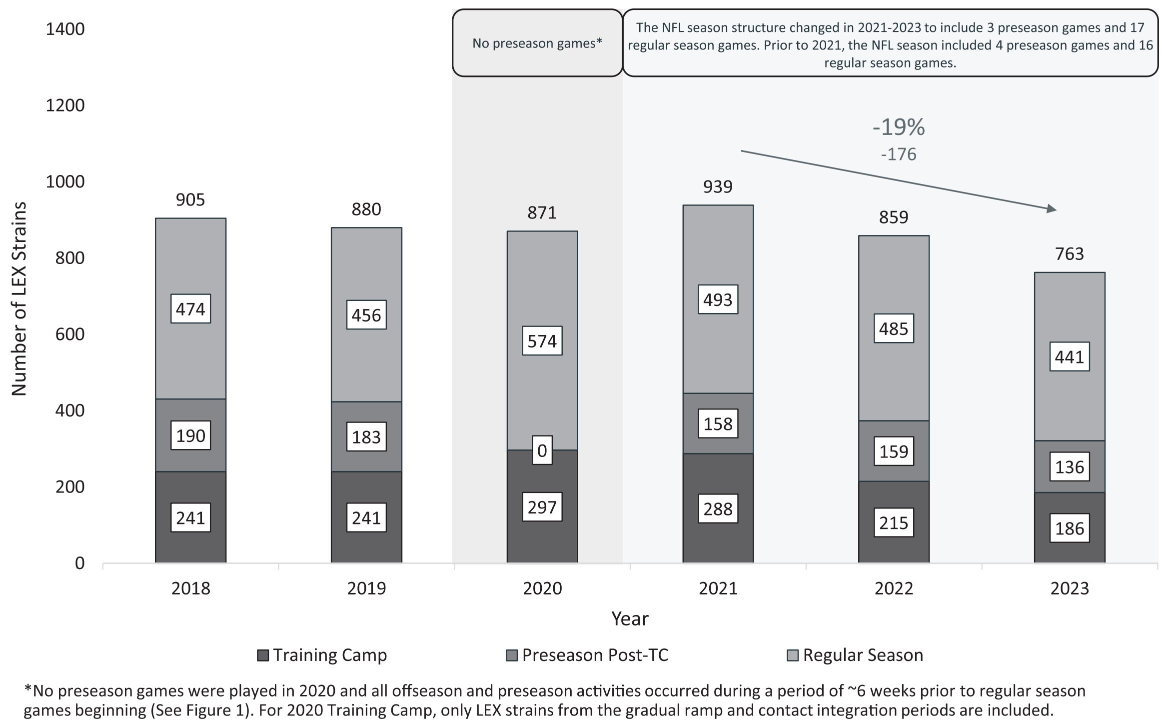


Figure 7. Preseason and regular season lower extremity (LEX) strains in games, practices, and other activities, 2018-2023.

when a new stimulus (eg, contact) was introduced, a reduction in load and intensity would be similarly required to allow players to acclimate to the new demands.

The concept of load management has been an increasing point of focus in sports settings over the past decade.[3,6,18,28] The recommendation to use a gradual acclimation strategy in the first week of training camp aligns with application of periodized training, which is an established training principle aimed at maximizing training to improve athletic performance by adjusting duration, load, and intensity of training over varying periods of time.[19,25,29] A 2001 study of strength and conditioning practices within the NFL identified that 69% of strength and conditioning coaches surveyed reported following a periodization model.[5] Although that study occurred approximately 20 years before the current one, this study found that before the mandate, a similar percentage of 67% of included club-seasons (55 of 82 club-seasons) appeared to use a periodization-type strategy in the first week of training camp in the form of a gradual duration ramp or high-low duration variation, although a <10 minute ramp preintervention was most frequently used (27 of 55 club-seasons).

Despite the popularity of training load management and periodized training, a paucity of data-driven evidence is available to support the effectiveness of specific strategies in reducing injuries. The few existing studies have largely evaluated individual training load monitoring practices[6,12,18] rather than population-based training programs. However, the results of the current study are similar to a previous study evaluating a linear periodization program in the military that found a low injury rate with a linear periodization program, although the subsequent effects of the program on future injury risk were not evaluated.[14] Regardless of the setting, these data, along with expert opinion, support considering gradual acclimation strategies when starting or returning to sports participation, including across other sports and other levels of sport.

Although duration, or time, is only a proxy for managing load, there are benefits to using time to prescribe load management. First, time is easy to communicate to coaches and their interdisciplinary medical and performance teams. Second, a time limit is easier to monitor than other load metrics (eg, distance, accelerations), which require equipment and dedicated personnel to collect, review, and interpret. Although these tools are available in professional sports settings, they are not universally accessible in other levels of sport. Third, implementing a time limit allows flexibility for the coaching staff and athletes to design their training sessions. Clubs used various strategies for designing practice periods after the mandate despite being required to do so within a set duration.

Given that time is a proxy, and potentially a coarse proxy, for monitoring load, it is imperfect. Other metrics, such as intensity of practice, physical fitness, and chronic load exposure, should also be considered. In this study, 66% of clubs had a lower median number of LEX strains after the mandated intervention. Some clubs that followed the gradual practice acclimation ramp saw no difference or slight increases in LEX strain and/or noncontact injury incidence, which could be due to various factors such as the type or volume of activities performed outside of structured team practices, the type or intensity of activities performed during the allotted practice time, other factors that influence injury occurrence (eg, player age, injury history, conditioning), or simply chance.

A key contributor to the success of the mandated gradual practice acclimation ramp in the NFL is believed to be the synchronized individual club education sessions. Information about the mandate was provided during these interdisciplinary meetings, inclusive of front office staff, coaches, and medical and performance staff. Importantly, key concepts related to the "spirit" of the mandate were emphasized, such as the importance of individually monitoring player internal and external load, comprehensively reviewing player load including nonpractice activities, achieving gamelike intensity in practice once players are acclimated, and allowing time for recovery as needed.

Limitations

First, the analysis does not identify causal factors explaining the influence of preseason practice schedule factors on injury occurrence. Additional factors may have influenced injury trends over time that were not considered in this analysis. Second, there is potential for exposure misclassification at the individual level; the strategy of the club may not reflect the player acclimation experience. Additionally, individual players may have had different levels of acclimation before the start of training camp. Third, potential exists for outcome misclassification, despite significant efforts to standardize EMR reporting across all 32 NFL clubs.[4] Additionally, evaluating additional injury outcomes beyond LEX strains and noncontact injuries was outside of the scope of this study. Fourth, results may not generalize to other levels of American football or other sports. In addition, 24% of wearable club-seasons were excluded from the analysis due to missing or incomplete practice information in training camp week 1, and there was limited wearable player tracking data available in the 2018-2019 seasons because use of a wearable device and provision of data were voluntary. Sensitivity analyses were performed of these club-seasons, and missingness was not found to affect findings. Fifth, the ideal way to prepare an athlete for sports participation from a performance perspective while limiting the risks of injury requires continued research to broadly understand how to optimize player health and performance.

CONCLUSION

A mandated gradual duration ramp in training camp coupled with education around strategic load management was associated with reduced LEX strains in the NFL. After the intervention, league-wide LEX strain incidence during training camp decreased by 35%.

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